

PRODUCT REQUIREMENTS DOCUMENT

Halcyon Credit — Agentic Underwriting Copilot

Version 2.0 · Sprint 0 · Futureense AI Clinic · Capstone Project 02

1 Executive Summary

Halcyon Credit is a digital consumer lender offering personal loans to applicants who frequently hold thin or non-traditional credit histories. Applications arrive faster than underwriters can work them manually. Each file requires income verification, bureau cross-checks, policy validation, and a documented decision rationale. The queue grows; handle time is high; decisions vary across reviewers; and decline reasons are written ad-hoc with no traceability to specific policy clauses.

This PRD specifies a human-in-the-loop Agentic Underwriting Copilot that assembles the facts of an application, predicts repayment risk using a trained ML model, evaluates compliance against the applicable loan-scheme policy via a Retrieval-Augmented Generation (RAG) engine, verifies documents and regulatory signals, audits the decision for fairness, and produces an evidence-backed recommendation. The system recommends — it does not decide. Final authority remains with a licensed human underwriter.

2 Problem Statement

Current underwriting at Halcyon Credit exhibits five measurable failure modes:

- **Time intensity.** Average file review time is approximately 45 minutes per application. The copilot target is ≤ 22 minutes ($\geq 50\%$ reduction).
- **Scalability.** Application volume during surges exceeds underwriter capacity, creating backlogs that damage applicant experience and revenue.
- **Inconsistency.** Same-profile applications receive different outcomes across reviewers. Decision consistency target: $\geq 95\%$ on matched profiles.
- **Traceability gap.** Decline reasons are written ad-hoc and not linked to policy clauses, creating regulatory exposure. Audit readiness target: 100%.
- **Cost.** Manual underwriting is expensive to scale. Unit cost target: $< \$0.05$ per application with the copilot.

A pure automation solution is unsuitable. Lending decisions require explainability, regulatory defensibility, policy adherence, fairness monitoring, and human accountability. The requirement is a copilot.

3 Vision Statement

VISION Enable underwriters to make faster, more consistent, and more explainable lending decisions through a transparent AI-assisted workflow that combines ML risk prediction, RAG-grounded policy evaluation, document

verification, regulatory validation, and fairness auditing — with a human underwriter in the loop for every final decision.

4 Target Users & Jobs to Be Done

4.1 Loan Underwriter (Primary)

Goal: assess applications accurately and efficiently, with evidence assembled for them.

- Large application queues with repetitive manual cross-checks
- Inconsistent decision documentation, especially on borderline cases
- Difficulty writing defensible decline reasons tied to specific policy clauses

Job to be done: "When reviewing an application, give me the applicant's verified risk profile, the applicable policy findings, and an evidence-backed draft recommendation — so I can approve, adjust, or override without re-reading the raw file."

4.2 Compliance Officer (Secondary)

Goal: ensure every decision is traceable, policy-adherent, and defensible to regulators.

- Lack of audit trails linking decisions to specific policy clauses or applicant data
- Unclear basis for decisions on edge cases or borderline profiles

Job to be done: "When auditing a decision, show me why the recommendation was produced, which policy clauses apply, whether fairness requirements were satisfied, and what the human underwriter changed — so I can defend it to a regulator."

4.3 Operations Manager (Tertiary)

Goal: maintain throughput, control cost, and monitor quality without increasing headcount.

- Underwriting backlogs and capacity constraints during volume surges
- Opaque cost and quality metrics with no per-application unit economics

Job to be done: "When monitoring underwriting operations, show me throughput, per-application cost, latency, recommendation acceptance rate, and escalation rate — so I can identify bottlenecks and control spend."

5 Scope

Status	Item
IN SCOPE	Probability of default prediction using supervised ML on structured loan application data
IN SCOPE	Multi-scheme policy evaluation — Personal Loan, Education Loan, Vehicle Loan, Business Loan, First-Time Borrower Program, Low-Income Assistance Program
IN SCOPE	Retrieval-Augmented Generation (RAG) policy engine grounding every recommendation in retrieved policy clauses
IN SCOPE	Policy version management and scheme-aware recommendation generation
IN SCOPE	Fairness monitoring across protected and proxy demographic segments on every decision

Status	Item
IN SCOPE	Evidence-backed recommendation generation (Approve / Decline / Refer) with cited applicant data, risk score, and policy references
IN SCOPE	Human-in-the-loop workflow with defined escalation triggers
IN SCOPE	Audit logging: every decision stores inputs, model outputs, policy checks, fairness results, and final recommendation
IN SCOPE	Retrieval quality monitoring (context precision, context recall, faithfulness)
IN SCOPE	Underwriter productivity and cost-per-application monitoring
OUT OF SCOPE	Autonomous loan approval or decline without human sign-off
OUT OF SCOPE	Live credit bureau API integrations (mocked in v1)
OUT OF SCOPE	Real identity verification or KYC integrations (mocked in v1)
OUT OF SCOPE	Production-grade financial deployment or regulatory licensing
OUT OF SCOPE	Applicant-facing portal or communication channels
OUT OF SCOPE	Fraud detection beyond signals available in the dataset
OUT OF SCOPE	Multi-jurisdiction legal compliance beyond simulated policy corpus
OUT OF SCOPE	Loss Given Default (LGD) and Exposure at Default (EAD) modelling — recovery rate estimation, collateral valuation, and collections assumptions are portfolio risk functions operated post-disbursement by the Risk Analytics team. Even though they can influence credit decisions rarely, its kept as a future extension in V2

6 Functional Requirements

ID	Requirement	Description	Outputs
FR-1	Application Ingestion	Ingest and normalise structured loan application data from upstream systems.	Normalised applicant profile
FR-2	Credit Risk Prediction	Estimate probability of default (PD) using a trained ML model along with top contributor	Risk score · Risk band · top contributor features

ID	Requirement	Description	Outputs
		features for the score.ML model — not the LLM — produces the risk number.	
FR-3	Multi-Scheme Policy Evaluation	Evaluate the application against the applicable loan-scheme policy (Personal, Education, Vehicle, Business, First-Time Borrower, Low-Income). Policy clauses must be retrieved via RAG before evaluation. Support policy versioning and a complete audit trail of applied clauses.	Applicable policy set · Passed rules · Failed rules · Escalation requirements
FR-4	Policy Retrieval Engine	Retrieve policy clauses from the lending policy knowledge base before recommendation generation. Every recommendation must reference retrieved evidence — no recommendation without supporting retrieval context.	Retrieved policy passages · Retrieval confidence score · Source attribution
FR-5	Document Verification	Verify uploaded applicant documents (salary slips, bank statements, employment letters, identity and address proofs, explanation letters): completeness, consistency, cross-document validation, and fraud signal detection.	Verification findings · Consistency findings · Missing information report
FR-6	Regulatory Validation	Validate applicant information against mock regulatory services (identity, employment, tax, sanctions). Retry policy and human-escalation fallback must be implemented. The system must never fabricate a regulatory response.	Verification status · Compliance findings · Escalation trigger if unresolved
FR-7	Fairness Audit	Evaluate every decision for disparate impact across protected and proxy demographic groups. Raise a bias alert and trigger escalation if the fairness gap exceeds the guardrail threshold.	Fairness indicators · Bias alerts · Segment-level approval-rate delta
FR-8	Recommendation Generation	Generate one of three recommendations — Approve, Decline, or Refer to Human Review — supported by risk score, policy findings, regulatory checks, document findings, and fairness result. The system may not emit Approve or Decline without a complete evidence chain.	Recommendation · Evidence chain · Escalation flag
FR-9	Explanation Generation	Produce a human-readable, auditable rationale grounded in retrieved policy clauses, applicant data, and ML model outputs. Every factual claim in the explanation must be traceable to a source.	Narrative explanation · Policy references · Supporting applicant data · Risk model factors
FR-10	Audit Logging	Record a complete, immutable decision record for every application processed.	Inputs · ML outputs · Policy checks · Regulatory checks · Document findings · Fairness result · Final recommendation · Timestamp
FR-11	Human-in-the-Loop Workflow	Route applications meeting any escalation criterion to a human underwriter queue. Escalation triggers: missing information, ML confidence below threshold, policy conflict, regulatory verification failure, fairness alert, thin-file flag.	Escalation record · Reason code · Underwriter queue entry

7 Non-Functional Requirements

Dimension	Requirement
Latency	the time until the system returns a complete, auditable recommendation ≤ 10 s. Maximum (P95) ≤ 20 s. ML inference budget ≤ 300 ms. RAG retrieval budget ≤ 600 ms.
Availability	Target uptime ≥ 99% during pilot operations.
Cost ceiling	Maximum cost per assessed application < \$0.05. Guardrail alarm triggers human-path fallback if exceeded.
Data privacy	No raw PII in LLM prompts. Application identifier and aggregated features only. SSN, DOB, and account numbers redacted before any model call.
Explainability	Every recommendation must contain an evidence chain: applicant data references, ML risk factors (SHAP), retrieved policy clauses (with source IDs), regulatory check result, fairness result.
Audit trail	Every decision record is immutable and queryable. Compliance officer must be able to reconstruct the full reasoning path from stored artefacts alone.
Human fallback	Any failure in ML inference, RAG retrieval, document verification, or regulatory check must degrade to human-review path. The system must never auto-decide on a failed sub-component.
Security	API keys managed via secrets manager, never in code or prompts. Least-privilege tool access. Audit logging enabled for all external service calls.

8 Success Metrics & KPIs

NORTH STAR: weighted Business Decision Utility Score combining AUC (weight 0.3), Policy Adherence (0.25), Recommendation Acceptance Rate (0.25), and Fairness Compliance pass/fail gate (0.20) .

GUARDRAIL 1 Fairness gap across demographic segments must never exceed 5 percentage points. This is a hard ship block — no headline metric improvement overrides it.

GUARDRAIL 2 Cost per assessed application must never exceed \$0.08. If breached, the system degrades to the human-review path automatically.

GUARDRAIL 3 Policy adherence must reach 100%. Any recommendation that violates a stated lending rule is a ship-blocking defect.

Supporting KPIs with targets and do-not-ship floors:

Category	KPI	Definition	How measured	Target	Do-not-ship floor
ML Quality	Decision quality (AUC)	Agreement with ground-truth outcome on held-out loans	AUC on held-out test set	≥ 0.80	< 0.70
ML Quality	Precision	True positives / (TP + FP) on default prediction	Sklearn classification report	≥ 75%	< 65%
ML Quality	Recall	True positives / (TP + FN) on default prediction	Sklearn classification report	≥ 70%	< 60%
RAG Quality	Policy retrieval accuracy	% of applications where the correct loan-scheme policy is retrieved	Manual eval test scenarios	≥ 95%	< 85% → spike fails
RAG Quality	Context precision	% of retrieved policy chunks that are relevant to the query	RAGAS context precision	≥ 85%	< 70%
RAG Quality	Context recall	% of relevant policy clauses successfully retrieved	RAGAS context recall	≥ 85%	< 70%
RAG Quality	Retrieval failure rate	% of requests where no relevant policy context is retrieved	System log count	< 2%	> 5% → ship block
Explanation	Faithfulness	Generated explanation grounded in retrieved evidence and applicant data	RAGAS faithfulness + LLM judge rubric	≥ 90%	< 75%
Explanation	Answer relevancy	Explanation answers underwriting question using retrieved evidence	RAGAS answer relevancy	≥ 90%	< 75%
Explanation	Hallucination rate	% of explanations containing claims unsupported by applicant data or retrieved policy	LLM judge + human spot-check	< 1%	> 3% → ship block
Fairness	Fairness gap	Approval rate delta across protected/proxy demographic segments	Equal opportunity difference on test set	≤ 3 pp	> 5 pp → HARD BLOCK
Fairness	Escalation accuracy	% of high-risk and thin-file applications correctly routed to human review	Labelled escalation set	≥ 90%	< 80%
Operations	Policy adherence	% of recommendations consistent with applicable lending policy rules	Rule-based policy checker + LLM judge	100%	< 98% → ship block
Operations	Cost per application	LLM token cost + embedding cost per assessed application	Gateway token log aggregated per run	< \$0.05	> \$0.08 → guardrail
Operations	P95 latency	End-to-end assessment time at 95th percentile	Load test at 50 concurrent requests	≤ 20 s	> 30 s
Operations	Audit readiness	% of decisions with complete evidence chain stored and queryable	Audit log completeness check	100%	< 99% → ship block

Category	KPI	Definition	How measured	Target	Do-not-ship floor
Business	Recommendation acceptance rate	% of copilot recommendations accepted by underwriters without substantive edit	Pilot: 3 underwriters × 20 files	≥ 75%	< 50%
Business	Review time reduction	Reduction in average underwriter time per file vs. baseline (45 min)	Time-on-task measurement in pilot	≥ 50% (≤ 22 min)	< 20% reduction
Business	Human override rate	% of recommendations rejected or substantively edited by underwriters	Pilot log	< 25%	> 40%

9 Risks & Initial Mitigations

This is a high-level risk identification for the PRD stage.

Risk	Likelihood	Impact	Mitigation / acceptance criterion
★ Policy RAG miss — wrong or no policy clause retrieved for the decision scenario	High	Critical	De-risk spike targets this assumption in Sprint 0. Do-not-ship floor: retrieval accuracy < 85%.
Biased ML model — disparate impact on thin-file or demographic segments	High	Critical	Fairness gap guardrail (> 5 pp) is a hard ship block. Segment-level metrics measured every sprint.
LLM hallucinates a policy rule not present in retrieved context	Medium	Critical	Every policy claim must cite a retrieved chunk source ID. Hallucination rate floor: > 3% blocks ship.
Over-automation of edge cases — low-confidence or thin-file applications auto-decided	Medium	High	Hard escalation rule: ML confidence < 0.6 or thin-file flag → human-review path, no auto-decision.
Cost blowout — LLM context grows through multi-step workflow	Medium	Medium	Per-call token budget enforced at gateway. Cheap model tier for non-reasoning steps. Guardrail > \$0.08 triggers human path.
Regulatory mock API failure causes unresolved verification	Low	High	Retry policy with exponential back-off; fallback to secondary mock; human escalation if unresolved. Never fabricate response.
Inconsistent policy versioning — stale policy clauses retrieved	Low	High	Policy knowledge base versioned and dated. Retrieval returns source version metadata on every call.

★ Policy RAG miss is the riskiest assumption.

10 Assumptions & Open Questions

#	Assumption	Status	Comment
A-1	Historical repayment outcomes (ground-truth labels) are available in the chosen dataset.	Confirmed	HC 2018 application_train.csv contains the TARGET column fully populated across all 307,511 rows (0 = repaid on time, 1 = payment difficulty). Zero missing values. Ground-truth label is available as-is for ML training and held-out evaluation.
A-2	A synthetic lending policy rulebook exists and can be used as clean text for RAG indexing.	Confirmed	A synthetic lending policy rulebook of 15–25 clauses will be authored for the pilot, covering DTI thresholds, income minimums, loan-to-income limits, thin-file escalation rules, documentation requirements, and fairness guardrails. This corpus is the RAG retrieval target for the de-risk spike and Sprint 1.
A-3	Human underwriters remain the final decision authority — no autonomous approve/decline in v1.	Confirmed	Explicitly stated in the capstone brief and locked in PRD Section 11 (Human-in-the-Loop Policy). The system generates recommendations only. No downstream disbursement or loan management action is triggered by system output.
A-4	Home Credit Default Risk 2018 is representative enough of the thin-file borrower population for pilot experimentation.	Confirmed	HC 2018 was built explicitly for financial inclusion and underbanked borrower populations, directly matching Halcyon Credit's stated problem. Confirmed as primary dataset for pilot scope. Caveat: Geographic and temporal distribution may differ from Halcyon's real applicant pool. Real-world deployment requires retraining on proprietary data. This caveat will be documented in the dataset card.
A-5	Bureau data is available as structured fields — no unstructured bureau narrative in v1.	Confirmed	HC 2018 bureau.csv and bureau_balance.csv are entirely structured — credit type, outstanding amounts, days-past-due history, status codes (C/X/O–5), and monthly balance history. No unstructured narrative text exists in either bureau table. Mock regulatory API calls in v1 will return structured JSON responses consistent with this schema.
A-6	Third-party LLM API is acceptable; on-premise inference is not required for the pilot.	Confirmed	Brief permits third-party APIs and free cloud tiers but PII masking layer is required
A-7	Pilot load target is 50 concurrent requests; peak production volume is assumed at < 200 applications per hour.	Assumed	Halcyon Credit is fictional — no real volume data is available. The < 200 applications per hour figure and 50 concurrent requests load test target are design parameters chosen for the pilot.
A-8 a	Protected and proxy demographic attributes required for fairness monitoring are present in the dataset.	Confirmed	HC 2018 contains the following attributes confirmed in the dataset schema: CODE_GENDER (M/F/XNA), NAME_EDUCATION_TYPE (5 levels), NAME_FAMILY_STATUS (6 values), DAYS_BIRTH (age proxy), REGION_RATING_CLIENT (socioeconomic proxy). Fairness evaluation across these segments is fully supported without additional data collection.
A-8 b	CODE_GENDER and age (DAYS_BIRTH) are excluded from ML training features and retained	Open — confirm before Sprint 1	CODE_GENDER_M has a positive correlation with default ($r = 0.054$) in HC 2018. Including it as an ML training feature may constitute direct discrimination. Design intent is to exclude CODE_GENDER and

#	Assumption	Status	Comment
	exclusively for fairness monitoring.		DAYS_BIRTH from ML features entirely and retain them only for fairness audit and segment-level evaluation. This must be confirmed as the agreed approach before Sprint 1 feature engineering begins. Note: Proxy leakage through correlated features (OCCUPATION_TYPE, REGION_RATING_CLIENT) must also be assessed in the fairness audit sprint.
A-9	Lending policy rulebook update cadence is monthly or slower; re-indexing is triggered manually on document change in v1.	Assumed	Since Halcyon Credit is fictional and the policy corpus is authored synthetically, update cadence is a design decision. Assumed cadence for pilot: monthly. Re-index of the Chroma vector store is triggered manually

11 Human-in-the-Loop Policy

The system may recommend, explain, verify, and escalate. It may not autonomously approve or decline a loan application. Final authority rests with a licensed human underwriter.

Mandatory escalation triggers

- ML model confidence score < 0.60 or thin-file flag raised
- Missing or conflicting applicant documents that cannot be resolved automatically
- Any policy rule violation or policy conflict
- Regulatory verification failure after retry policy exhausted
- Fairness alert raised on the decision
- Any sub-component failure (ML, RAG, document verification, regulatory check)

KILL SWITCH If any guardrail metric is breached, or if the system cannot produce a complete evidence chain, it must immediately degrade to the human-review path and raise an operational alert. It must never guess, fabricate, or auto-decide.

12 Acceptance Criteria

The project is accepted when all of the following are satisfied in the final Sprint 4 evaluation:

ID	Acceptance criterion
AC-1	End-to-end underwriting workflow (ingestion → risk score → policy check → fairness audit → recommendation → audit log) completes without error on test cases.
AC-2	ML model AUC ≥ 0.80 on held-out test set. F1 ≥ 0.72.
AC-3	Policy retrieval accuracy ≥ 95% across all supported loan schemes.
AC-4	Policy adherence reaches 100% — no recommendation violates a stated lending rule.

ID	Acceptance criterion
AC-5	Fairness gap ≤ 5 pp across demographic segments on test set (hard block if exceeded).
AC-6	Every recommendation contains a complete evidence chain: risk factors, policy clause citations, regulatory check results, document findings.
AC-7	Hallucination rate $< 1\%$ measured by LLM judge on held-out explanation set.
AC-8	Cost per assessed application $< \$0.05$ under representative load.
AC-9	P95 end-to-end latency ≤ 20 s at 50 concurrent requests.
AC-10	Human underwriter can audit every recommendation from stored artefacts alone (audit readiness = 100%).
AC-11	Recommendation acceptance rate $\geq 75\%$ in underwriter pilot (3 underwriters $\times$ 20 files).

End of PRD v2.0 · Halcyon Credit Agentic Underwriting Copilot · Futureense AI Clinic Capstone Project 02